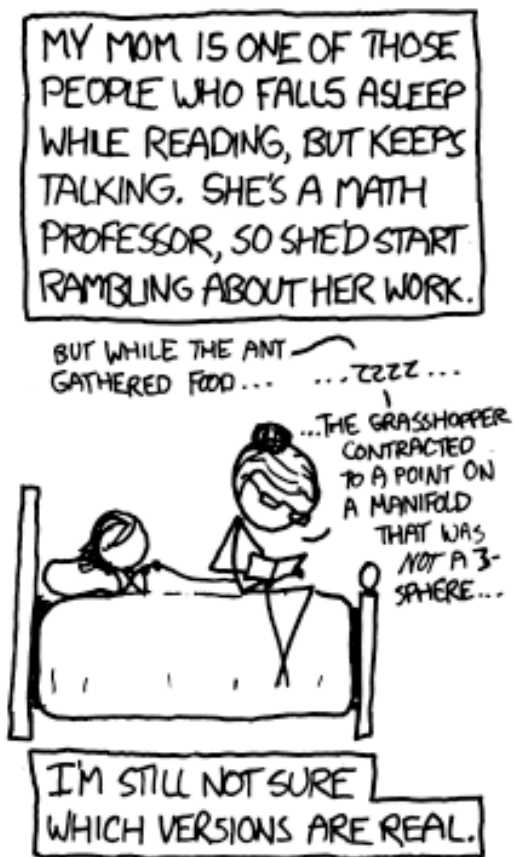
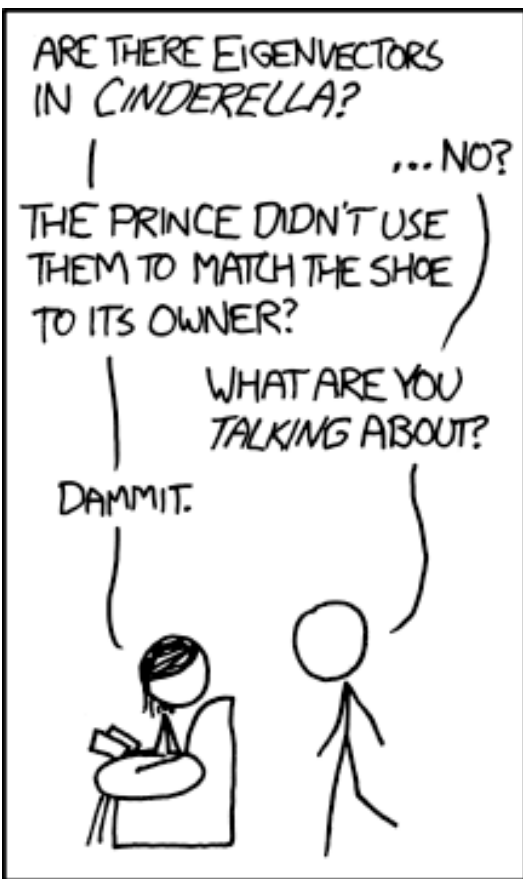




<https://xkcd.com/872/>

Classic Mechs EP2: Recap of some essential math



COE COLLEGE

Classic Mechs EP2: Recap of some essential math

Learning objectives

- 1) We will learn/review the basic rules of matrix multiplication, dot product.
- 2) We will cover two applications:
 - Linear transformation & coordinate transformation
 - Derive state-space from Newtonian mech to pave the way to...
solve matrix systems of equation with eigenvalues and eigenvectors

```
import numpy as np
np.dot (A,x)  or  A @ x
```

Matrix multiplication, dot product

2x2 example:

$$M3 = M1 \times M2 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \bullet \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 \times a_2 + b_1 \times c_2 & a_1 \times b_2 + b_1 \times d_2 \\ c_1 \times a_2 + d_1 \times c_2 & c_1 \times b_2 + d_1 \times d_2 \end{bmatrix}$$

Generalization:

$$P_{i,j} = \sum_{k=1}^n Q_{i,k} \bullet R_{k,j}$$

$$P = \begin{bmatrix} Q_{11}R_{11} + Q_{12}R_{21} + \cdots + Q_{1n}R_{n1} & Q_{11}R_{12} + Q_{12}R_{22} + \cdots + Q_{1n}R_{n2} & \cdots & Q_{11}R_{1q} + Q_{12}R_{2q} + \cdots + Q_{1n}R_{nq} \\ Q_{21}R_{11} + Q_{22}R_{21} + \cdots + Q_{2n}R_{n1} & Q_{21}R_{12} + Q_{22}R_{22} + \cdots + Q_{2n}R_{n2} & \cdots & Q_{21}R_{1q} + Q_{22}R_{2q} + \cdots + Q_{2n}R_{nq} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{m1}R_{11} + Q_{m2}R_{21} + \cdots + Q_{mn}R_{n1} & Q_{m1}R_{12} + Q_{m2}R_{22} + \cdots + Q_{mn}R_{n2} & \cdots & Q_{m1}R_{1q} + Q_{m2}R_{2q} + \cdots + Q_{mn}R_{nq} \end{bmatrix}$$

Matrix element indexing: [i, k]

Row #

Column #

The dot product application: linear transformation and coordinate transformation

Unfortunately, no one can be told what the Matrix is. You have to see it for yourself.

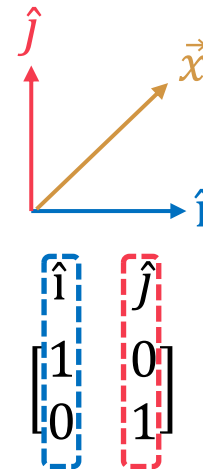
-Morpheus

<https://youtu.be/kYB8lZa5AuE?si=su9SausqkmoxiRoU>

$\vec{x}$: Every single vector in space

$$\begin{matrix} \vec{x'} & \text{Output vector} \\ \vec{x} & \text{Input vector} \end{matrix}$$
$$\begin{bmatrix} \\ \end{bmatrix} = \begin{bmatrix} & \end{bmatrix} \begin{bmatrix} \\ \end{bmatrix}$$

A matrix



Recall: any vector can be described with basis vectors $\hat{i}$ and $\hat{j}$.

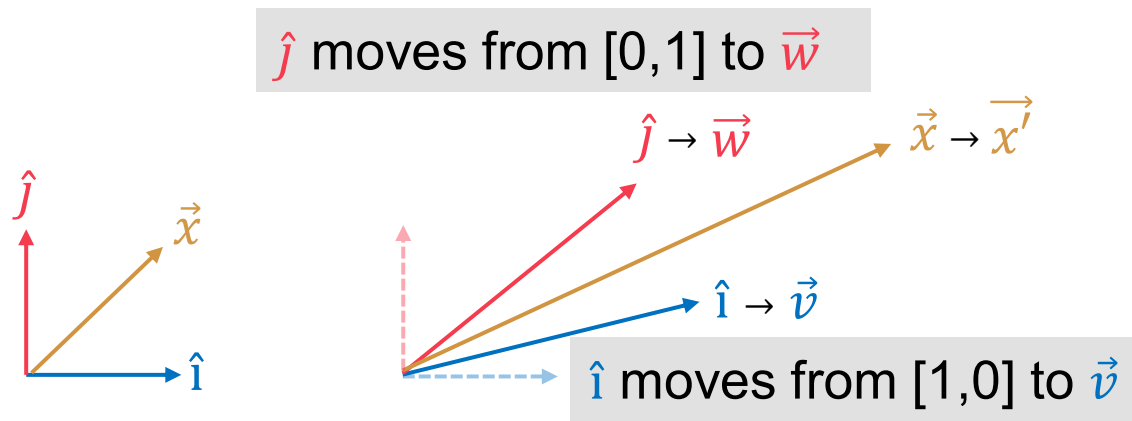
How do we describe a linear transformation mathematically without tracking every single point in space?

The dot product application: linear transformation and coordinate transformation

Unfortunately, no one can be told what the Matrix is. You have to see it for yourself.

-Morpheus

<https://youtu.be/kYB8lZa5AuE?si=su9SausqkmoxiRoU>



$$\begin{bmatrix} \hat{i} \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} \hat{j} \\ 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} \vec{v} \\ \vec{w} \end{bmatrix}$$

A matrix

$$\underline{x'} = \underline{A} \underline{x}$$

The dot product application: linear transformation and coordinate transformation

A 90° Counter-Clockwise Rotation:

- $\hat{i}$ moves from $[1, 0]$ to $[0, 1]$.
- $\hat{j}$ moves from $[0, 1]$ to $[-1, 0]$.
- The resulting rotation matrix is: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

A Shear Transformation:

- $\hat{i}$ remains fixed at $[1, 0]$.
- $\hat{j}$ leans over to $[1, 1]$.
- The resulting shear matrix is: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Also Euler rotations:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation of $\hat{i}$ Rotation of $\hat{j}$



More example of coordinate transformation:

Sequential Euler rotations in 3D: $\mathbf{v}_{final} = \mathbf{R}_z(\psi) \cdot \mathbf{R}_x(\theta) \cdot \mathbf{R}_z(\phi) \cdot \mathbf{v}_{initial}$

$$\mathbf{R}_z(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{R}_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \quad \mathbf{R}_x(\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{bmatrix}$$

Might be an interesting read: loss of degree of freedom (DOF) and gimbal lock: https://en.wikipedia.org/wiki/Gimbal_lock.

Bonus: describe kinetic energy (T) in metric tensor format

2D Cartesian Coordinates (x, y)

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$$

↓

$$T = \frac{1}{2}m \begin{bmatrix} \dot{x} & \dot{y} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix}$$

metric tensor g

2D Polar Coordinates (r, θ)

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$$

↓

$$T = \frac{1}{2}m \begin{bmatrix} \dot{r} & \dot{\theta} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & r^2 \end{bmatrix} \begin{bmatrix} \dot{r} \\ \dot{\theta} \end{bmatrix}$$

3D Cylindrical Coordinates (r, θ, z)

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2)$$

↓

$$T = \frac{1}{2}m \begin{bmatrix} \dot{r} & \dot{\theta} & \dot{z} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{r} \\ \dot{\theta} \\ \dot{z} \end{bmatrix}$$

This is how we get metric tensor: $g = J^T J$,

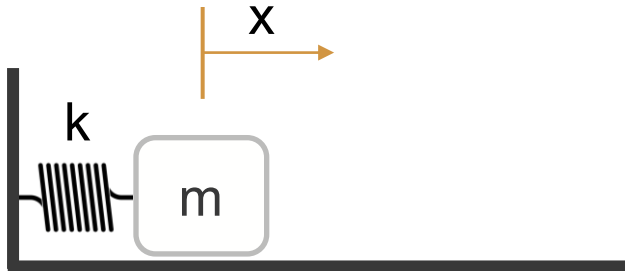
where $J = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix}$



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The dot product application: state-space

Single harmonic oscillators*:



2nd order linear DE

$$\ddot{x} = -\frac{k}{m}x$$



A system of coupled
1st order linear DEs

$$\dot{x} = v$$

$$\dot{v} = -\frac{k}{m}x$$

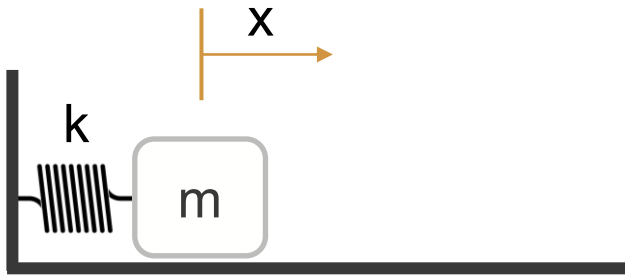
$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$

WHY?

https://youtube.com/shorts/AqME_OqamWQ

The dot product application: state-space

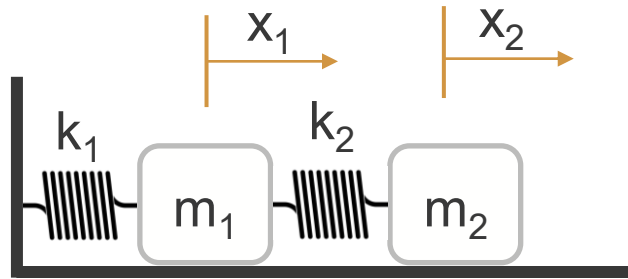
Single harmonic oscillators: Double harmonic oscillators case A: Double harmonic oscillators case B:



$$\dot{x} = v$$

$$\dot{v} = -\frac{k}{m}x$$

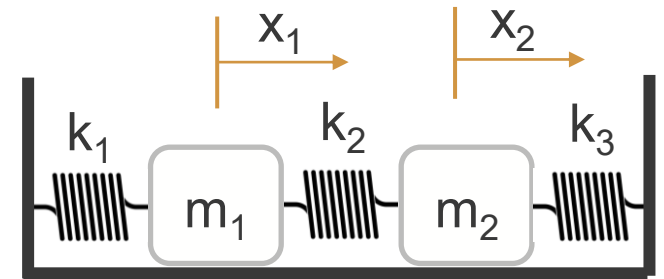
$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$



$$m_1 \ddot{x}_1 = -k_1 x_1 + k_2 (x_2 - x_1)$$

$$m_2 \ddot{x}_2 = -k_2 (x_2 - x_1)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{v}_1 \\ \dot{x}_2 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & 0 & \frac{k_2}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & 0 & -\frac{k_2}{m_2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{bmatrix}$$



*Assuming: no friction, no air resistance